DEFENSE AI INFRASTRUCTURE ENGINEERING

Kenny Sheridan

Infrastructure Product Engineer for AI, robotics, edge, agentic workflows, and supercomputing systems.

Systems engineer with 10+ years of experience turning complex infrastructure into production-ready platforms.

I build and evaluate reproducible AI, robotics, edge, and datacenter systems for agentic workloads, including bare-metal GPU orchestration, Kubernetes control planes, distributed Raft-backed reconciliation loops, high-performance networking and storage, QEMU/KVM sandboxing, observability, benchmarking, workload-to-hardware matching, and technical stack/infrastructure viability assessments.

My experience spans medium-sized companies scaling into Fortune 500 operations, as well as pre-seed through post-Series A startups valued from $300M to $1B. I work well in startup environments where priorities change quickly and ideas need to become shipped systems without losing reliability or technical depth. I'm a former U.S. Marine Corps meteorology instructor with hands-on HPC delivery experience across NVIDIA and AMD platforms for AI/ML, robotics, edge, simulation, and datacenter workloads.

9,000+
GPUs automated across heterogeneous providers and clusters

800G+
InfiniBand and NVMe-oF fabric experience

10+ yrs
Production infrastructure and product monetization

LOCATION *Seattle, WA* EMAIL *kennethdashensheridan@gmail.com* GITHUB *github.com/kennethdsheridan* FORGEJO *forgejo.kennysheridan.io*
LINKEDIN *linkedin.com/in/kennethdashensheridan*

EXPERIENCE

Member of Technical Staff - Infrastructure Product | Andromeda

Greater Seattle Area, Remote | March 2026 - Present | Website: andromeda.ai

- Build production AI infrastructure reliably matched to model training and inference workloads.
- Automate observability and deployment for 9,000+ GPUs across over a dozen globally distributed providers and clusters supporting training and inference capacity for AI/ML, robotics, edge, and datacenter workloads.
- Develop bare-metal GPU orchestration, hardware lifecycle workflows, distributed control planes, and developer-facing platforms for maintainable infrastructure product delivery.
- Use Nix to build reproducible, secure package and deployment environments; use QEMU/KVM for sandboxing, isolation, and repeatable infrastructure validation.
- Create rich product documentation for customer-facing infrastructure workflows, designed for online hosting, mobile viewing, and clear operational adoption.
- Design virtualization and multi-tenant compute isolation patterns that make large-scale GPU infrastructure safer, more repeatable, and commercially usable.
- Drive performance engineering, longitudinal system modeling, cache strategy, storage evaluation, offload analysis, and workload placement so the right models land on the right hardware and serve quickly.

Senior Supercomputing Infrastructure Engineer | San Francisco Compute Company

Greater Seattle Area, Remote | September 2024 - March 2026 | Website: sfcompute.com

- Led automated bring-up for 2,000 NVIDIA H100 GPUs, moving bare metal into operational Kubernetes clusters through a single-command Rust-based deployment workflow.
- Scaled onboarding from 8 nodes to hundreds of GPU nodes within weeks by eliminating manual provisioning steps across hardware, networking, and company infrastructure integration.
- Deployed distributed supercomputing infrastructure globally for GPU marketplace capacity, with emphasis on scalable utilization, reliability, and operational repeatability.
- Optimized compute, SDN, network fabric, and high-performance storage for large-scale AI and HPC workloads.
- Built performance testing suites and custom Kubernetes controllers/operators for bare-metal clusters, resource management, near-metal validation, and repeatable systems setup.

Senior AI and HPC Infrastructure Engineer | TensorWave

Greater Seattle Area, Remote | May 2024 - July 2024 | Website: tensorwave.com

- Architected on-premises AMD MI300X GPU clusters using EPYC CPUs and RDMA over Converged Ethernet on traditional TCP/IP networks.
- Benchmarked NVIDIA InfiniBand and AMD RoCE designs, including high-bandwidth all_reduce testing over 800G switching infrastructure.

SKILLS

CODE	Linux, Rust, Go, Bash
BUILD	Nix packaging, QEMU/KVM
CONTROL	Kubernetes operators, GitOps, Argo CD
AGENTIC	Workflows, harnesses, MCP
DATA	Raft reconciliation, embedded databases
HARDWARE	Bare metal, Redfish, H100, H200, B200, B300, MI300X
GPU	CUDA, ROCm, NCCL
FABRIC	RoCE, InfiniBand, NVMe-oF, SONiC, SDN
OVERLAY	Tailscale, Netmaker
OFFLOAD	SmartNICs, Smart Storage cards
STORAGE	Weka, VAST, Ceph
TELEMETRY	OpenTelemetry, Prometheus, Grafana
PERF	FIO, iperf3, stress-ng, MPI, PyTorch

STRENGTHS

Defense AI infrastructure
Build platforms for AI/ML, robotics, simulation, edge, and datacenter workloads.

Reproducible systems
Make infrastructure repeatable, auditable, and safe to operate across environments.

GPU and edge orchestration
Automate accelerator fleets across providers, clusters, and constrained networks.

Fabric and storage paths
Design data movement, storage access, offload paths, and telemetry for reliable workloads.

Validation-first delivery
Benchmark, stress, profile, model, and place workloads with operational evidence.

OPEN SOURCE

- Designed vendor-agnostic AI/ML infrastructure patterns capable of scaling toward hundreds of nodes while reducing accelerator lock-in.
- Created deployment documentation for GPU cluster setup, configuration, and operational handoff.

Senior Hardware Infrastructure Automation Engineer | ServiceNow

Kirkland, WA | February 2022 - December 2023 | Website: servicenow.com

- Engineered HPC system testing software for distributed enterprise infrastructure, including stress validation, benchmarking, and reliability assessment workflows.
- Led migration of internal automation from Python and Bash to Go, improving efficiency across heterogeneous hardware environments.
- Built Redfish-based SKU auditing, NIC benchmarking, and GitLab CI/CD workflows for hardware-software validation.

Senior Cloud Hardware Performance Test Engineer | ServiceNow

Kirkland, WA | May 2017 - February 2022 | Website: servicenow.com

- Led hardware performance testing across storage, networking, BIOS, firmware, PCIe, FPGAs, SmartNICs, Smart Storage cards, NVMe, Linux filesystems, Weka, VAST, and Ceph.
- Worked with ODMs, system engineers, CTO stakeholders, and product teams to refine infrastructure roadmaps and train engineers on repeatable test methods.

System Administrator | NexLevel Information Technology

Sacramento, CA | August 2015 - May 2017 | Website: nexlevelit.com

- Provided Tier 3 Unix and Windows server support for biometric systems serving 300+ remote clients, including storage recovery, monitoring scripts, and production baseline improvements.

Technical Instructor of Meteorology | U.S. Marine Corps

Quantico, VA | May 2007 - June 2015 | Website: marines.mil

- Administered two modular data centers, maintained METMF(R) computing infrastructure, virtualized instructional environments, and managed WAN-connected remote sensing sites.
- Produced 300+ surface observations, 100+ forecasts, and 50+ weather warnings cited in Navy and Marine Corps Achievement Medal recognition.

SELECTED ENGINEERING WORK

Repeatable AI infrastructure environments

Nix | QEMU/KVM | Secure packaging | Model-serving cache paths

Set up deterministic infrastructure environments that use Nix for secure packages, QEMU/KVM for sandboxed validation, and caching strategies to serve AI models quickly.

Multi-cluster HPC observability platform

Rust | Prometheus | Grafana | HPC workloads | Infrastructure monitoring

Custom-built single-pane-of-glass observability system for multi-cluster HPC workloads and infrastructure, designed to integrate with Prometheus and Grafana workflows.

Automated GPU bring-up and onboarding

Rust | Kubernetes | Bare metal | NVIDIA H100

Single-command workflow that deploys hardware, joins company infrastructure, configures networking, and removes manual intervention from large-scale GPU node onboarding.

Multi-node GPU cluster networking

AMD MI300X | NVIDIA | RoCE | InfiniBand | 800G fabrics

Designed vendor-agnostic topologies across AMD and NVIDIA accelerators, including RoCE on TCP/IP networks and InfiniBand benchmarking for large-scale AI/HPC clusters.

LLM and HPC benchmarking CLI

Rust | MPI | PyTorch | DeepSeek-R1 | NVIDIA and AMD GPUs

Resource-efficient benchmarking application for rapid system characterization across InfiniBand, LLM workloads, and heterogeneous GPU infrastructure.

Forgejo portfolio

Self-hosted forge for systems architecture and private/public engineering work.

GitHub profile

Public Rust, Nix, GPU, MCP, and developer tooling repositories.

Neovim tooling

Lua-based personal and professional editor configuration used since 2015.

EDUCATION

Air University / Community College of the Air Force

Keesler AFB | 62 credits in meteorology, forecasting, and atmospheric physics

CERTIFICATIONS

Introduction to Blockchain

Amazon Web Services, 2021

Basic Instructor Course

Community College of the Air Force, 2012

Meteorology and Oceanography Analyst Forecaster

Community College of the Air Force, 2008

RECOGNITION

Navy and Marine Corps Achievement Medal

Good Conduct Medal

Letter of Appreciation, Director of Meteorology